A certified counterexample to complete monotonicity

MathDB 351688: the Mills-ratio derivative quotient

Principia Math In Progress

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Task 1. Resolve the stated complete-monotonicity conjecture

Let

$$r(z) = e^{z^2/2} \int_z^\infty e^{-t^2/2}\, dt, \qquad F_n(z) = -\frac{(r^{(n)}(z))^2}{r^{(2n+1)}(z)}.$$

Conjecture 3 in Ferreira–Simon [1, p. 31] asserts that every F_n is completely monotone on $(0,\infty)$. The definition of r is on p. 29. Complete monotonicity requires $(-1)^k F_n^{(k)}(z) \geq 0$ for every nonnegative integer k and $z > 0$. The following counterexample uses $n = 1$.

Theorem 1. *For the exact positive argument $c = 1/100$,*

$$-\frac{461}{10^{54}} < \frac{F_1^{(70)}(c)}{70!} < -\frac{460}{10^{54}} < 0. \tag{1}$$

Consequently, F_1 is not completely monotone, and the universal assertion in Conjecture 3 is false.

Publication scope. The complete 33-page author manuscript [1] has now been read. The authors announced this version as accepted for publication in Transactions of the American Mathematical Society. The source audit uses this accepted manuscript; the separately typeset journal version remains unread. The package records the bounded claim audit.

Proof. Changing variables in the defining integral yields

$$r(z) = \int_0^\infty e^{-zt-t^2/2}\, dt, \qquad r'(z) = zr(z) - 1.$$

Differentiation under the integral is valid locally uniformly for real z, because the Gaussian dominates every polynomial times a fixed exponential. In particular,

$$-r'''(z) = \int_0^\infty t^3 e^{-zt-t^2/2}\, dt > 0.$$

Thus the quotient and all its derivatives are well defined on the required interval.

Put $R(w) = r(c-w) = \sum_{k\geq 0} e_k w^k$. The differential equation for r gives the exact coefficient recurrence

$$e_0 = r(c), \qquad e_1 = 1 - ce_0, \qquad e_{k+1} = \frac{e_{k-1} - ce_k}{k+1} \quad (k \geq 1). \tag{2}$$

Moreover,

$$F_1(c-w) = \frac{R'(w)^2}{R'''(w)} = \sum_{k\geq 0} q_k w^k, \qquad q_k = \frac{(-1)^k F_1^{(k)}(c)}{k!}.$$

Define

$$a_k = (k+1)e_{k+1}, \qquad b_k = (k+1)(k+2)(k+3)e_{k+3}.$$

Since $b_0 = -r'''(c) > 0$, coefficient comparison gives

$$q_k = \frac{1}{b_0}\left(\sum_{j=0}^{k} a_j a_{k-j} - \sum_{j=1}^{k} b_j q_{k-j}\right). \tag{3}$$

Computing through e_{73} and q_{70} therefore determines the required derivative without numerical differentiation.

For completeness, we specify the exact finite certificate for its sign. Every bound below uses integers and rational numbers. Set

$$A(v) = \sum_{j=0}^{199} \frac{(-1)^j v^{2j+1}}{2j+1}, \qquad T(v) = \frac{v^{401}}{401}.$$

The alternating-series estimate gives $A(v) < \arctan v < A(v) + T(v)$ for $v = 1/5, 1/239$. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ then gives explicit rational lower and upper bounds $\pi_- < \pi < \pi_+$. The identity follows directly from the tangent addition formula; the angle $4\arctan(1/5) - \arctan(1/239)$ lies in $(0, \pi/2)$ and has tangent one.

Let $S = 10^{200}$ and

$$h = \left\lfloor S\sqrt{\pi_-/2} \right\rfloor.$$

Integer comparisons verify

$$(h/S)^2 < \pi_-/2 < \pi_+/2 < ((h+1)/S)^2.$$

Consequently $h/S < \sqrt{\pi/2} < (h+1)/S$. To enclose $e_0 = r(c)$, use

$$r(c) = e^{c^2/2}\left(\sqrt{\pi/2} - \int_0^c e^{-t^2/2}\,dt\right).$$

For $u = c^2/2 = 1/20000$, define

$$E = \sum_{j=0}^{79} \frac{u^j}{j!}, \qquad J = \sum_{j=0}^{79} \frac{(-1)^j c^{2j+1}}{2^j j!(2j+1)}.$$

Positive-series and alternating-series bounds give

$$E < e^u < E + \frac{u^{80}}{80!}\frac{1}{1-u/81},$$
$$J < \int_0^c e^{-t^2/2}\,dt < J + \frac{c^{161}}{2^{80}80!\,161}.$$

These enclose the sole nonrational input e_0.

Propagate these intervals through (2) and (3), rounding outwards after every operation to the grid $S^{-1}\mathbb{Z}$. For multiplication take the minimum and maximum of the four endpoint products; for division take the four endpoint quotients, having checked that the denominator interval is strictly positive. Round lower endpoints down and upper endpoints up. Induction on the recurrences proves that every computed interval contains the exact coefficient. The accompanying standard-library Python verifier performs precisely these operations, using arbitrary-length integers. Its final interval is strictly inside

$$\left(-\frac{461}{10^{54}}, -\frac{460}{10^{54}}\right).$$

This is (1), because the derivative order 70 is even. The exact rational endpoints are saved in `exact-certificate.json`. This finite certificate proves a counterexample to a universal statement; it is not a finite-sample argument for an infinite assertion. □

The same verifier additionally certifies

$$-\frac{302}{10^{28}} < \frac{F_0^{(40)}(0)}{40!} < -\frac{301}{10^{28}}, \qquad -\frac{758}{10^{54}} < \frac{F_1^{(70)}(0)}{70!} < -\frac{757}{10^{54}}.$$

Continuity would already give violations at small positive arguments. The theorem instead supplies the explicit point $1/100$. This result does not contradict Proposition 10 of [1], which concerns the different functions $z \mapsto F_n(\sqrt{z})$.

An analytic explanation of the failure

The failure also has a proof that does not depend on the coefficient computation. In fact, at every fixed positive argument there are infinitely many derivative orders with the wrong sign.

Write $h(z) = -r'(z) = 1 - zr(z)$ and

$$H(z) = -r'''(z) = \int_0^\infty t^3 e^{-zt-t^2/2}\,dt = (z^2+3)h(z) - 1.$$

Both functions are entire, and $H(x) > 0$ for every real x. Moreover,

$$|H(z)| \le 8e^{|z|^2}, \qquad x^4 H(x) = \int_0^\infty s^3 e^{-s-s^2/(2x^2)}\,ds \longrightarrow 6 \quad (x \to +\infty).$$

The growth bound follows from $|z|t \le |z|^2 + t^2/4$; the limit follows by dominated convergence.

Suppose H had no zeros. It would have an entire logarithm g, with $H = e^g$ and $\Re g(z) \le \log 8 + |z|^2$. For the Taylor coefficients $g(z) = \sum d_j z^j$, the Fourier coefficient formula on the circle of radius R, applied to $\log 8 + R^2 - \Re g \ge 0$, gives

$$|d_j| R^j \le 2(\log 8 + R^2 - \Re g(0)), \qquad j \ge 1.$$

Letting $R \to \infty$ shows that g has degree at most two. But $\log|H(x)| = \Re g(x)$ would then be a real polynomial of degree at most two, whereas the displayed limit gives $\log H(x) = -4\log x + \log 6 + o(1)$. This is impossible. Hence H has a nonreal zero. At every zero of H, the identity $H = (z^2+3)h - 1$ implies $h \neq 0$. Thus $F_1 = h^2/H$ has a nonreal pole and has no real poles.

We recall a short form of the positive-coefficient argument. If $G(w) = \sum a_j w^j$, $a_j \ge 0$, has finite positive radius of convergence ρ, it cannot be analytic at $w = \rho$. Indeed, if it were, monotone convergence as $w \uparrow \rho$ would give

$$\frac{G^{(k)}(\rho)}{k!} = \sum_{j \ge k} \binom{j}{k} a_j \rho^{j-k}.$$

The Taylor series at ρ converges at $\rho + \varepsilon$ for some $\varepsilon > 0$. Interchanging its nonnegative sums then gives $\sum_j a_j(\rho+\varepsilon)^j < \infty$, a contradiction.

For any fixed $c > 0$, the Taylor series of $F_1(c - w)$ has finite positive radius because of its nonreal pole, but the function is analytic at every real w. Its coefficients therefore cannot all be nonnegative. Nor can they be eventually nonnegative: subtracting finitely many Taylor terms would give the same contradiction. Since these coefficients are $(-1)^k F_1^{(k)}(c)/k!$, infinitely many have the wrong sign. This proves non-complete-monotonicity analytically, as well as explaining why a high derivative can detect it.

Tasks 2–5. Proposed directions after the counterexample

These are proposed mathematical tasks, not additional solved claims.

Task 2. Determine the smallest order k for which $(-1)^k F_n^{(k)}(z) < 0$ somewhere on $(0,\infty)$, first for $n = 0, 1$, and then its dependence on n.

Task 3. Decide whether every F_n, $n \geq 0$, fails complete monotonicity. Extend the question to real moment orders by replacing $(-1)^n r^{(n)}(z)$ with $\int_0^\infty t^\nu e^{-zt-t^2/2}\,dt$, $\nu > -1$.

Task 4. Characterize the powers $\beta > 0$ for which $z \mapsto F_n(z^\beta)$ is completely monotone. In particular, determine whether the known exponent $\beta = 1/2$ is sharp.

Task 5. Locate the nearest complex zeros of $r^{(2n+1)}$, identify cancellations in F_n, and use the resulting singularities to derive asymptotics and sign changes of its high derivatives.

References

[1] R. A. C. Ferreira and T. Simon, *Convolution of beta prime distribution*, arXiv:2108.09244v2 (19 May 2022), Section 5.5, Conjecture 3, https://arxiv.org/abs/2108.09244v2. Journal metadata: Trans. Amer. Math. Soc. 376 (2023), 855–890, https://doi.org/10.1090/tran/8748. The author-announced accepted manuscript was read completely; the separately typeset version of record was not accessible.